

■ 3.9.5 Numerical Solution of Polynomial Equations

When `Solve` cannot find explicit solutions to polynomial equations, it returns a symbolic form of the result.

```
In[1]:= Solve[x^5 + 7x + 1 == 0, x]
Out[1]= {ToRules[Roots[7 x + x^5 == -1, x]]}
```

You can get numerical solutions by applying `N`.

```
In[2]:= N[%]
Out[2]= {{x -> -1.11308 - 1.15173 I},
          {x -> -1.11308 + 1.15173 I}, {x -> -0.142849},
          {x -> 1.1845 - 1.15139 I}, {x -> 1.1845 + 1.15139 I}}
```

This gives the numerical solutions to 16-digit precision.

```
In[3]:= N[%, 16]
Out[3]= {{x -> -1.11307797654771 - 1.151734362151675 I},
          {x -> -1.11307797654771 + 1.151734362151675 I},
          {x -> -0.1428486455250044},
          {x -> 1.184502299310213 - 1.151390075408837 I},
          {x -> 1.184502299310213 + 1.151390075408837 I}}
```

You can use `NRoots` to get numerical solutions to polynomial equations directly, without first trying to find exact results.

```
In[4]:= NRoots[x^7 + x + 1 == 0, x]
Out[4]= x == -0.796544 || x == -0.705298 - 0.637624 I ||
          x == -0.705298 + 0.637624 I ||
          x == 0.123762 - 1.05665 I ||
          x == 0.123762 + 1.05665 I ||
          x == 0.979808 - 0.516677 I ||
          x == 0.979808 + 0.516677 I
```

This converts the solutions to a sequence of transformation rules.

```
In[5]:= {ToRules[%]}
Out[5]= {{x -> -0.796544}, {x -> -0.705298 - 0.637624 I},
          {x -> -0.705298 + 0.637624 I},
          {x -> 0.123762 - 1.05665 I},
          {x -> 0.123762 + 1.05665 I},
          {x -> 0.979808 - 0.516677 I},
          {x -> 0.979808 + 0.516677 I}}
```

<code>NRoots[poly==0, x]</code>	get approximate numerical solutions to a polynomial equation
<code>NRoots[poly==0, x, n]</code>	get solutions to n digit precision
<code>{ToRules[eqns]}</code>	write solutions from <code>Roots</code> or <code>NRoots</code> in the form of transformation rules

Numerical solution of polynomial equations.

`NRoots` will always give you the complete set of numerical solutions to any polynomial equation in one variable.

You can also get numerical solutions to sets of simultaneous polynomial equations. You can use `Solve` to “unwind” the simultaneous equations, and then apply `N` to get numerical results.

`Solve` writes these equations as a composition of simpler equations.

```
In[6]:= Solve[{x^2 + y^2 == 1, x^3 + y^3 == 2}, {x, y}]
Out[6]= {ToRules[Roots[3 x ==
      y (-3 - 6 y - y^2 + 4 y^3 + 2 y^4), x,
Using ->
      Roots[3 y^2 - 4 y^3 - 3 y^4 + 2 y^6 == -3, y]]]}
```

You can apply `N` to get a numerical result.

```
In[7]:= N[%]
Out[7]= {{x -> -1.09791 + 0.839887 I,
      y -> -1.09791 - 0.839887 I},
{x -> -1.09791 - 0.839887 I,
      y -> -1.09791 + 0.839887 I},
{x -> 1.22333 - 0.0729987 I,
      y -> -0.125423 - 0.712005 I},
{x -> 1.22333 + 0.0729987 I,
      y -> -0.125423 + 0.712005 I},
{x -> -0.125423 - 0.712005 I,
      y -> 1.22333 - 0.0729987 I},
{x -> -0.125423 + 0.712005 I,
      y -> 1.22333 + 0.0729987 I}}
```